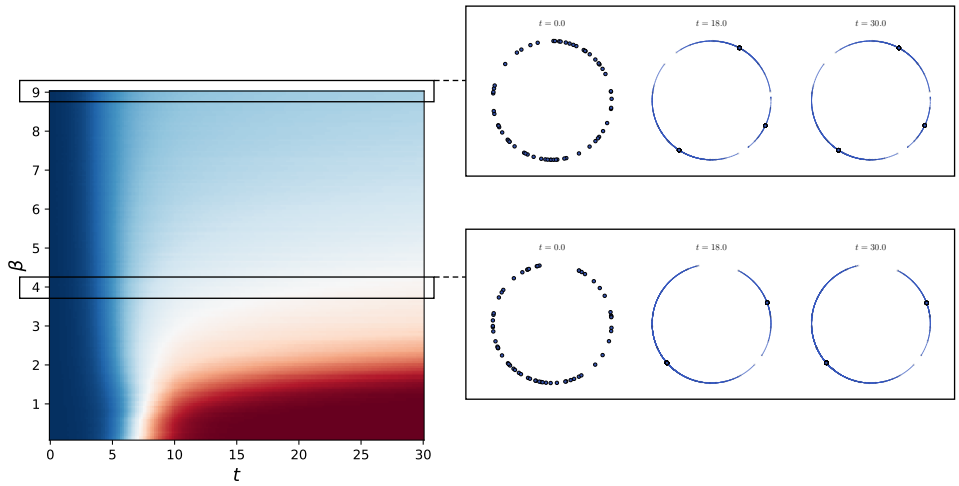


Figure 4. We zoom in on the phase diagram (Figure 3) for the dynamics on the circle: $d = 2$. For $\beta = 4, 9$, we also display a trajectory of (SA) for a randomly drawn initial condition at times $t = 2.5, 18, 30$. We see that the particles settle at 2 clusters when $\beta = 4$ (bottom right) and 3 clusters when $\beta = 9$ (top right), for a duration of time. This reflects our metastability claim for large β.

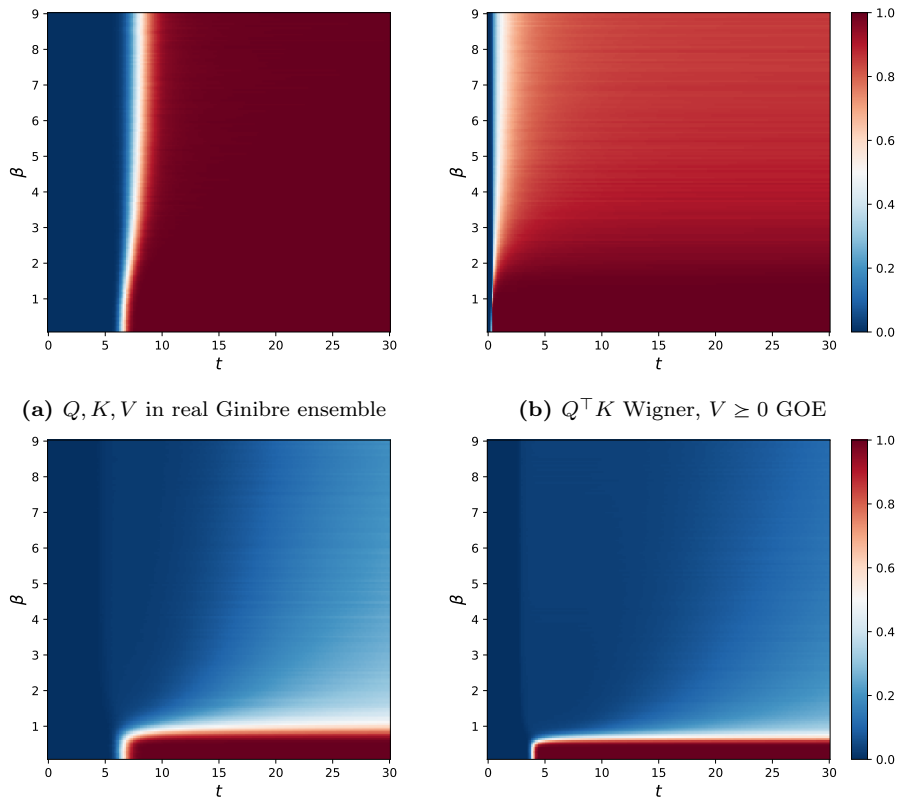
- In Section 8, we discuss an alternative approach for analyzing clustering, based on the so-called *BBGKY hierarchy* from statistical mechanics;
- In Section 9, we foray beyond the case $Q^\top K = V = I_d$ in a few directions. We start with Section 9.1 by relating the case $V = -I_d$ to *optimal configurations* on the sphere. We then discuss existing results in the absence of layer normalization in Section 9.2, which motivate the study of a related singular equation (Section 9.3) and a diffusive regularization (Section 9.4);
- Finally, in Section 10, we briefly discuss various ways focusing on the tuning of the parameter matrices themselves.

7. Dynamics on the circle

We study the dynamics (SA) and (USA) in the special case $d = 2$, namely on the unit circle $\mathbb{S}^1 \subset \mathbb{R}^2$. This model, parametrized by angles and related to the celebrated Kuramoto model, is of independent interest and deserves a complete mathematical analysis.

7.1. **Angular equations.** On the circle $\mathbb{S}^1$, all particles $x_i(t) \in \mathbb{S}^1$ are of course completely characterized by the angle $\theta_i(t) \in \mathbb{T}$: $x_i(t) = \cos(\theta_i(t))e_1 + \sin(\theta_i(t))e_2$ where $e_1 = (1, 0)$ and $e_2 = (0, 1) \in \mathbb{R}^2$. We focus on the dynamics (USA) for simplicity. For any $i \in [n]$ and $t \geqslant 0$, we may derive the equation satisfied by $\theta_i(t)$ from $\cos(\theta_i(t)) = \langle x_i(t), e_1\rangle$: differentiating in t and plugging into (USA) we obtain

$$\dot{\theta}_i(t) = -\frac{n^{-1}}{\sin(\theta_i(t))}\left(\sum_{j=1}^{n} e^{\beta\langle x_i(t), x_j(t)\rangle}\Big[\langle x_j(t), e_1\rangle - \langle x_i(t), x_j(t)\rangle\langle x_i(t), e_1\rangle\Big]\right)$$



(a) Q, K, V in real Ginibre ensemble

(b) $Q^\top K$ Wigner, $V \geq 0$ GOE

(c) Q, K in real Ginibre ensemble, $V = Q^\top K$

(d) $Q^\top K$ Wigner, $V = Q^\top K$

Figure 5. Phase diagrams (see Figure 3 for explanations) for some choices of random matrices (Q, K, V); here $d = 128$, $n = 32$. Sharp phase transitions as well as metastable regions appear in all cases.

where we used the definition of the projection (if $\theta_i(t) = 0$ for some t, we differentiate the equality $\sin(\theta_i(t)) = \langle x_i(t), e_2 \rangle$ instead, which also leads to (7.1) in the end). Observing that

$$\langle x_i(t), x_j(t) \rangle = \cos(\theta_i(t) - \theta_j(t)),$$

we find

$$\dot{\theta}_i(t) = -\frac{n^{-1}}{\sin(\theta_i(t))} \left(\sum_{j=1}^{n} e^{\beta \cos(\theta_i(t) - \theta_j(t))} \Big[\cos(\theta_j(t)) - \cos(\theta_i(t) - \theta_j(t)) \cos(\theta_i(t)) \Big] \right).$$

Using elementary trigonometry, we conclude that

$$\dot{\theta}_i(t) = -\frac{1}{n} \sum_{j=1}^{n} e^{\beta \cos(\theta_i(t) - \theta_j(t))} \sin(\theta_i(t) - \theta_j(t)). \tag{7.1}$$

The case $\beta = 0$ is exactly the Kuramoto model recalled in Section 7.2. Suppose for the time being that $\beta > 0$. Defining the function $h_\beta : \mathbb{T} \to \mathbb{R}_{\geqslant 0}$ as

$$h_\beta(\theta) = e^{\beta \cos(\theta)},$$

we have effectively deduced that the empirical measure of the angles, $\nu(t) = \frac{1}{n}\sum_{j=1}^{n}\delta_{\theta_j(t)}$, which is a measure on the torus $\mathbb{T}$, is a solution to the continuity equation

$$\partial_t \nu(t) + \partial_\theta(\mathcal{X}[\nu(t)]\nu(t)) = 0, \qquad \text{on } \mathbb{R}_{\geqslant 0} \times \mathbb{T},$$

where

$$\mathcal{X}[\nu](\theta) = \frac{1}{\beta}\Big(h'_\beta * \nu\Big)(\theta).$$

When the particles $x_i(t)$ follow (SA), one readily checks that the same continuity equation is satisfied but rather with the field

$$\mathcal{X}[\nu](\theta) = \frac{1}{\beta}\left(\frac{h'_\beta * \nu}{h_\beta * \nu}\right)(\theta).$$

7.2. **The Kuramoto model.** As mentioned above, when $\beta = 0$, (7.1) is a particular case of the Kuramoto model [Kur75]:

$$\dot{\theta}_i(t) = \omega_i + \frac{K}{n}\sum_{j=1}^{n}\sin(\theta_j(t) - \theta_i(t)), \tag{7.2}$$

where $K > 0$ is a prescribed coupling constant, and $\omega_i \in \mathbb{T}$ are the intrinsic natural frequencies of the oscillators $\theta_i(t)$. It is known that for sufficiently small coupling strength K, the oscillators $\theta_i(t)$ in the Kuramoto model (7.2) do not synchronize in long-time. It is also known that when K exceeds some critical threshold value, a phase transition occurs, leading to the synchronization of a fraction of the oscillators. If K is chosen very large, there is total synchronization of the oscillators in long-time. For more on the mathematical aspects of the Kuramoto model, we refer the reader to the review papers [Str00, ABV+05, HKPZ16] (see also [CCH+14, Chi15, FGVG16, DFGV18, HR20, TSS20, ABK+22, MP22] for a non-exhaustive list of other recent mathematical results on the subject).

When all the frequencies ω_i are equal to some given frequency, $\omega \in \mathbb{R}$ say, after a change of variable of the form $\theta_i(t) \leftarrow \theta_i(t) - \omega t$, the dynamics in (7.2) become the gradient flow

$$\dot{\theta}(t) = n\nabla \mathsf{F}(\theta)$$

where the energy $\mathsf{F} : \mathbb{T}^n \to \mathbb{R}_{\geqslant 0}$ reads

$$\mathsf{F}(\theta) = \frac{K}{2n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\cos(\theta_i - \theta_j). \tag{7.3}$$

The oscillators can be viewed as attempting to maximize this energy. The energy F is maximized when all the oscillators are synchronized, that is, $\theta_i = \theta^*$ for some $\theta^* \in \mathbb{T}$ and for all $i \in [n]$. As the dynamics follow a gradient system, the equilibrium states are the critical points of the energy, namely those satisfying $\nabla \mathsf{F}(\theta) = 0$. The local maxima of F correspond to equilibrium states θ that are physically achievable, since small perturbations thereof return the system back to θ.

Some authors consider a variant of the Kuramoto model where the oscillators are interacting according to the edges of a graph. In other words, the coefficients A_{ij} of the graph's adjacency matrix are inserted in the sum in (7.3) as weights, and the dynamics are then the corresponding gradient flow. A recent line of work culminating with [ABK+22] has established that synchronization occurs with high probability for Erdős–Rényi graphs with parameter p, for every p right above the connectivity threshold.